Prof. Dr.-Ing.
Timon Rabczuk

Fakultät Bauingenieurwesen

Institut für Strukturmechanik

Reference Letter



To whom it may concern,

I am pleased to write this letter in support of Mr. Yizheng Wang. Mr. Yizheng Wang has been visiting my computational mechanics group at the Institute of Structural Mechanics, Bauhaus University of Weimar, as a visiting PhD-student early 2024 for a 1 year visit and for also in January 2026. I met him beforehand several times in China where we started before our collaboration. Due to my close association with him for the past years, I know him very well both technically and personally. Mr. Wang's major focus in his doctoral studies is on AI for partial differential equations (AI4PDEs) with applications to challenging problems in engineering.

Mr. Yizheng Wang has an outstanding knowledge of continuum mechanics, modern numerical methods, machine learning approaches and their applications. He has the capability to implement new ideas extremely quickly and to prepare manuscripts on his work in a short time. His implementation skills are extraordinary and he knows many coding languages. Mr. Wang has developed several new computational methods like physics informed Kolmogorov Arnold Network, the (Complementary) Deep Energy Method based on different network architectures including techniques for operator learning and thoroughly verified and validated their capabilities.

He recently developed physics informed (or more precisely variational physics informed) neural operators which entirely avoid labeled data which dramatically improve the efficiency of current simulation workflows. To my best knowledge, he is the first to successfully develop those operators for varying geometries, boundary conditions and input parameters which is not only of tremendous academic but also industrial importance. He furthermore suggested the use of highly efficient initialization approaches for traditional solvers with further speed ups to a factor of 10 for specific problems.

Mr. Yizheng Wang is a very diligent and hard working student and his motivation is exceptional. He has implemented most of his methods from scratch. We have published jointly several manuscripts in reputable journals, including Nature Communications or Computer Methods

in Applied Mechanics and Engineering, and currently several manuscripts are in review. Mr. Wang is an exceptionally productive researcher and his academic capabilities are comparable already to assistant professor level.

Out of the 40 PhD students I graduated so far and the additional 120 PhD students and postdocs who visited my research lab over the past 20 years, Mr. Yizheng Wang clearly sticks out due to his outstanding technical and academic skills.

Mr. Wang is very organized and has very good administrative and organizational capabilities. He quickly adjusted to the new environment in Weimar and is well integrated in my team where he already has proven some leadership qualities helping and advising other PhD students in my group. He can quickly adopt to coordinate and work with several people in parallel. He is a congenial and very friendly coworker and has excellent relations with his colleagues. His commitment and the punctuality towards deadlines are commendable.

He has my strongest support and I am sure that he will make excellent contributions in the future.

Best regards,



Timon Rabczuk

Chair of Computational Mechanics

Prof. Dr.-Ing.
Timon Rabczuk

Fakultät Bauingenieurwesen

Institut für Strukturmechanik

Marienstraße 15
D-99423 Weimar

Postanschrift:
D-99421 Weimar

Telefon:
+49 (0) 36 43/58 45 11

Telefax:
+49 (0) 36 43/58 45 14

E-Mail:
timon.rabczuk@uni-weimar.de